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The benefit of clean water on child health: An empirical analysis with specific reference to *Escherichia Coli* water contamination

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ABSTRACT

Microorganism-mediated degradation of water quality is a major public health concern in developing countries. Previous literature has shown an association between household water pollution and childhood diarrhoea; however, its effects on child growth, respiratory health, and infant mortality have not been widely investigated. This study assesses the impact of household drinking water contaminated with *Escherichia coli (E. coli)* on child's weight-for-height and weight-for-age z-scores, acute respiratory infections (ARI), and diarrhoea incidence among five years children, and on infant mortality rate (IMR) in Pakistan. We use district-level geospatial information and the latest waves of unique Multiple Indicator Cluster Survey (MICS) data containing information on 'point-of-service delivery' (POS) and 'point-of-consumption' (POC) water quality, collected for the first time on a large scale in Pakistan. We employ an instrumental variable approach to address potential endogeneity issues in household drinking water quality, finding that POC drinking water contamination significantly affects children's weight-for-height and weight-for-age z-scores and diarrhoea but insignificantly affects ARI and IMR. The effects of contaminated water are found to be particularly significant in children older than 6 months of age and an insignificant effect is observed for younger children. To protect the children from growth failure and contracting diarrhoea, household water quality should be improved.

1. Introduction

Ending undernutrition, and reducing infectious illness and child mortality are critical prerequisites for sustainable development. Worldwide, around half of the under-five mortality (3 million) is attributable to undernutrition [1], which is linked to an amplified risk of morbidity and infections including respiratory illness [2]. Diarrhoea and acute respiratory infections (ARI) remain the leading causes of mortality and morbidity, especially in developing countries [3,4]. Every year, from one billion childhood diarrhoea episodes, around 5 million results in death [5] owing to infant and under-five mortality, and ARI contributes to 15 % of deaths under the age of five worldwide. The cumulative health burden is evident because diarrhoea can impair child growth [6] and trigger ARI [7].

Recent research and international development partners have focused on environmental contamination, a possible structural barricade to child development [5,8,9]. Among environmental risks related to health, water pollution in particular is considered an

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'invisible threat' to the global development agenda, and contaminated water can significantly increase morbidity and affect labour productivity and well-being [10]. E.coli bacterial contamination[1] is found to be one of the main causes of failure to meet the global criteria for safely managing drinking water [11]. Fecal exposure through poor drinking water, sanitation, and hygiene is linked to approximately 62 % diarrhoeal mortality, 13 % ARI burden, and 16 % malnutrition cases [12].

Water pollution, malnutrition, and infectious illness posing a threat to the lives of children are severe in Pakistan. The infant mortality rate (IMR) in the country is 53 per 1000 live births [13], one of the highest amongst the developing countries. Bacterial contamination is increasing due to rapid urbanisation, water scarcity, and poor sanitation. According to the WHO & UNICEF (2022) [14] access to safely managed drinking water is merely 35.8 %, indicating that even if the source water is safe, unhygienic storage and water handling may lead to contamination [15]. The *E. coli* presence indicates water exposure to animal or human faeces [16]. As the access to basic sanitation is 68.4 % in Pakistan, inadequate sanitation can elevate the risk of microbial contamination due to the extensive discharge of faecal sludge into water bodies [17,18]. The situation is exacerbated due to recent massive floods in the country which largely affected the water and sanitation systems and enhanced the risk of diarrhoea, ARI, and malnutrition in almost 10 million children [19].[2]

This study investigates the impact of *E. coli* contamination on weight-for-height and weight-for-age z-scores, ARI, and diarrhoea incidence in children up to 5 years of age, and infant mortality using Multiple Indicator Cluster Survey (MICS) data in Pakistan, which has one of the worst child health conditions in the world. MICS datasets have the unique feature of containing information on 'point of service delivery' (POS) and 'point-of-consumption' (POC) water quality, which was collected for the first time in a wide geographic area in Pakistan. This allows us to directly examine the impact of POC water quality on child health. Moreover, having information on POS enables us to identify community-level source water contamination. Since household drinking water is influenced by household decisions and behaviours, our model may suffer from a possible endogeneity problem. Some unobserved factors could be correlated with household water quality, which may bias our estimates Given the complexities of household behavior and unobserved factors, the direction of bias in the OLS estimates can indeed vary and manifest in either direction. To circumvent this issue, we use community-level POS contamination as an instrumental variable (IV) to account for the possible endogeneity of POC drinking water quality through empirical estimation.

Our results document that a household using contaminated drinking water has a higher probability of diarrhoeal episodes by 11 percentage points. We find that contaminated water is correlated with a decrease in children's weight-for-height by 0.41 and weight-for-age by 0.39 standard deviations. An insignificant and moderate effect of POC-contaminated water on the probability of ARI risk is observed. Based on these results, we infer that E. coli-induced episodes of diarrhoea could be a risk factor for undernutrition. In addition to diarrhoea, environmental enteric dysfunction (EED), a subclinical gut disorder triggered by relentless faecal-oral contamination, is a possible route linking water contamination with malnutrition [20,21], which might contribute to a weakened immune system. Lastly, we show an insignificant impact of POC-contaminated water on the infant mortality rate (IMR).

Furthermore, we explore the differential effects of water quality on child health, focusing on the age of the child. Dividing the sample by age group, we find that children older than 6 months are susceptible to *E. coli* contamination, possibly due to a larger intake of water and exposure to the contaminated environment than infants up to 6 months of age, who are generally exclusively or mostly breastfed. While exploring the breastfeeding channel, we find consistent and insignificant effects on the younger infants. One concern, however, could be the possibility of survival bias (i.e. weak children died due to poor quality of water). An insignificant effect of contaminated water on IMR in the main analysis could assuage this concern, indicating that survival bias is unlikely to be the major explanation for the insignificant effects on child health outcomes in the age-specific differential analysis.

We examine the differential impacts of water quality on child health depending on the provision of micronutrient supplements. We find that in the non-vitamin A-supplemented group, clean water access does not improve the child's growth, while it positively affects the reduction of diarrhoea. This suggests that while clean water access contributes to the reduction of diarrhoea, it does not help child growth possibly because Vitamin A deficiency is a bottleneck for a child's development. In contrast, we observe a significant effect of water quality on growth indicators for the vitamin A-supplemented group. It might be possible due to baseline vitamin A deficiencies, timing, or dosage of supplementation, underlying nutritional status, and pre-existing health conditions of children in this group. The clean water access and vitamin A supplementation might work as complements to improve children's growth and health outcomes. The effect of water quality remains insignificant on ARI for both groups.

Our study makes two major contributions to this field. The first contribution has epidemiological importance as it reveals the impact of *E. coli* contamination of POC water on ARI, child growth, and IMR. Although previous epidemiological studies have examined the impact of water quality on the incidence of diarrhoea [16,22–27] height-for-age Z score [27], and self-reported maternal health [28], the literature on the impact of *E. coli* contamination on ARI, child growth, and IMR is scant. Since the evidence suggests that water, sanitation, and hygiene (WASH) interventions and the choice of water sources affect child growth and ARI [29–35], it is likely the case that contaminated water can negatively affect child growth and ARI occurrence although some found no impact [35]. This is likely to be because of different nutritional status, immune systems, and different ages of observation. The impact of *E. coli* contamination on child health, however, has not been well studied due to the lack of data on microbial water quality. Previous studies

[1] *Escherichia coli (E. coli),* an indicator of faecal contamination is a species of faecal coliform group, easily cultured microorganisms and used to measure the water quality level. Certain strains of *E. coli* may develop resistance to multiple antimicrobial drugs (MDR), leading to MDR *E. coli* infections. MDR *E. coli* is indicated as a priority pathogen by WHO due to emerging antimicrobial resistance (AMR) [60].

[2] Inadequate WASH, mainly attributed to intestinal issues, typhoid, hepatitis, and respiratory illness, contributes to about 0.6–1.44 % of the burden on the country's GDP [61].

found a significant effect of surface water quality [9] and piped water coverage [36,37] on IMR. Greenstone & Hanna (2014) [38] focusing on river water quality found no significant relationship between water quality and IMR. However, we focus on the effects of POC microbial water contamination on IMR. Furthermore, we examine the differential impacts of *E. coli* contamination on child health by age differences and Vitamin A intake. In doing so, this study provides more detailed policy implications than previous studies did.

The second contribution of our study relates to our data set and identification strategy. As discussed, MICS datasets have the unique feature of containing information on POS and POC water quality. This data enables us to directly investigate the effectiveness of POC-contaminated water on child health by using community-level source water contamination (except self) as an instrument since POC drinking water quality could be endogenously determined. This identification strategy provides robust results and adds evidence to the literature, as a limitation of previous studies is that they largely ignored the endogeneity of household water quality. Although a few exceptions are [16,23], who also considered endogeneity issues and found that microbial water contamination significantly affects the incidence of diarrhoea, none of the other previous studies have done so.

The remainder of this paper is organised as follows. In Section 2, we explain the datasets and provide summary statistics. Section 3 delineates the empirical strategy. Section 4 documents the regression results of the analyses across several specifications, and Section 5 discusses the results. Finally, Section 6 concludes the paper, by summarizing the key findings and exploring their policy implications.

2. Data

2.1. Sampling

This study utilised data from the latest round of the Multiple Indicator Cluster Surveys (MICS) in Pakistan; a provincially representative cross-sectional survey followed a two-stage stratified random sampling technique. During the survey, rural and urban areas from each district were identified as the main sampling strata. Following two-stage stratified sampling, a specific number of Census Enumeration Areas (EA) were selected systematically with probability proportion to size within each stratum in the first stage [39]. Subsequently, a systematic sample of 20 households was selected from each sample EA (hereafter termed 'cluster/community'). We use the provincial MICS survey datasets of Pakistan conducted from 2017 to2021.[3]

A unique feature of MICS is the addition of a new water quality module developed by MICS and the Joint Monitoring Program (JMP) [40] which was released as a part of the latest round of MICS-6.[4] Worldwide, few datasets include bacteriological water quality indicators at a national scale owing to collection challenges, and in most cases, only the quality of piped water is measured. In our datasets, water quality data at the POC and POS were collected separately for the first time at a large scale in Pakistan using a universally accepted method [41].

In these MICS datasets, from 123,771 interviewed households 113,233 mothers/caretakers were interviewed for anthropometry and other health outcomes for children under five from study regions. For microbial water quality testing, 18,492 households were randomly selected. We limited our sample where households gave consent for the water test; enumerators visited the source for the *E. coli* test, and information for both the source and household water test was available. For the analyses, 13,603 observations remained for children aged 0–60 months and 5520 observations for infant mortality analysis after cleaning the data.

2.2. Outcome variables

This study mainly employs five outcome variables. First, we use diarrhoea prevalence, which takes the value of 1 if the child's mother/caregiver reported the child had diarrhoea two weeks before the survey. Second, ARI symptoms are used, which takes the value of 1 if the child had experienced rapid or difficult breathing along with a persistent cough and the symptoms were associated with chest problems and a blocked nose two weeks before the survey, excluding the children having only blocked nose [42,43].[5] The third and fourth outcomes are child growth: weight-for-height and weight-for-age z-scores, which are the standard deviations of the anthropometry of the child based on the growth standards of the WHO.[6] Lastly, we use the infant mortality rate (IMR) as an outcome; the infant mortality dummy takes 1 if the child did not survive to age one year and 0 otherwise. The IMR is scaled to the number of infant deaths per 1000 live births.

[3] Five survey waves; MICS Punjab 2017–18, MICS Sindh 2018–19, MICS KPK 2019, MICS Baluchistan 2019–20, and a region MICS AJK 2020–21, are combined. Although each survey, conducted in a different year, has the survey weight to make it provincial representative, we do not have the weight to make it nationally representative at a certain survey year. To complement the non-usage of sampling weight, we include an extensive set of controls in addition to time and geographic fixed effects to mitigate potential biases related to unobserved heterogeneity.

[4] Methods were developed for direct water testing in MICS. Generally, laboratories conduct water tests, which often suffer from logistic challenges in terms of transportation and time frame.

[5] Respiratory rate is a valuable clinical sign for diagnosing acute lower respiratory infections (e.g. pneumonia and bronchiolitis) in children coughing and breathing rapidly.

[6] WAZ and WHZ can be accumulatively affected by water quality, but these are more sensitive to recent variations in health and nutritional status. On the other hand, the children's height-for-age z-score (HAZ), an indicator of chronic malnutrition, is affected by long-term environmental exposure [5,27]. In MICS, water quality was tested at a certain point in time, and we lack long-term data on the water quality status. Hence, we do not include HAZ as an outcome variable in this study. Although the results (available upon request) are not shown, an insignificant effect of *E. coli* exposure on HAZ is observed.

2.3. Main explanatory variables

The key explanatory variable of POC-contaminated drinking water is a dummy variable, which takes 1 if *E. coli* was found in a household's POC water.[7] The World Health Organization's drinking water quality guidelines recommend that *E. coli* must not be detectable in 100-ml water samples [44]. A water test was performed on the samples provided by the households within 30 min of collection followed by 24–48 h of incubation. *E. coli* colonies were counted as colony-forming units (CFU)/100 ml of water sample. Since households use water from different sources, the source water sample was also collected for each household, and the *E. coli* contamination was calculated by using a similar method used for household water tests.[8]

Fig. 1 displays the specific *E. coli* count (CFU/100 ml) at POC in the left panel and at POS in the right panel. In the MICS datasets, the distribution of *E. coli* as a continuous indicator ranged from 0 to 101 CFU/100 ml of water, where all values above 100 are censored and coded as 101. We use this variable as a binary indicator in our main analysis because using it as a continuous indicator would be misleading. In our sample, the drinking water of around 79 % of households is found to be contaminated with *E. coli*, and the percentage of households having contaminated water at the source is around 61 %. The variation in *E. coli* concentration at the POC and POS shows that a higher *E. coli* risk (100 or above CFU/100 ml) is prevalent at POC than at POS, suggesting that, contamination occurs while fetching and handling drinking water.

In the regression analysis, we control for several children, mothers, and household characteristics. Child-level variables include gender and age. The mother and household characteristics comprise the age and education of the mother, the household head's gender, maternal smoking, household size, number of children up to 5 years of age, cooking place, number of household members, household assets, access to flush toilets as an indicator of improved sanitation,[9] livestock ownership, and survey year dummy. For the community-level variables, we include the averages of household open defecation behaviour, piped water facilities, household ownership, and land ownership for agriculture purposes at the community level as well as a rural dummy. Appendix Tables A1 and A2 present the summary statistics for the variables used in the analyses.

2.4. District characteristics

We supplement the information from the MICS with additional geospatial datasets to account for district-level characteristics that may affect health and water quality. We construct the district-level data on rainfall, population density, slope, and temperature. Based on the district's geocoded position, we link the MICS data to information on district-level characteristics. For rainfall data, we use the Climate Hazards Group InfraRed Precipitation with Station Data 2.0 (CHIRPS) [45]. For population density, we use WorldPop data [46]. For slope, we use ALOS DSM: Global 30m v3.2 data provided by the JAXA Earth Observation Research Centre (EORC). For temperature, we use the dataset provided by the NASA LP DAAC at the USGS EROS Centre [47]. To control for the economic status in each district, we use the VIIRS night-time light dataset constructed by NOAA's National Centres for Environmental Information [48]. In addition, we use the Malaria-Atlas map to obtain information on the average malaria prevalence rate in each district, as a proxy for malaria risk (*Plasmodium vivax* parasite rate in the general population) [49].

In Fig. 2, using district-level spatial information, we map child health data from the sample and plot the district-level risk prevalence of *E. coli* at POS and POC levels and health-related outcomes. The figure shows that the intensity of the prevalence of health issues is higher in some places where the *E. coli* concentration in POC water is comparatively higher.

3. Empirical strategy

This section describes the empirical strategy employed to estimate the impact of household POC water quality on children's health. Similar to Rahman et al. (2021) [23], we use Grossman's (1972) health production function approach [50] and estimate the model below focusing on two components: health inputs and outputs. The child's health outcome is assumed to be a function of the POC drinking water quality as a primary health input as well as other sociodemographic and environmental factors:

$$\mathbf{Y}_{ijkd} = \beta_0 + \beta_1 \mathrm{WQ}_{jkd} + \beta_2 \mathbf{H}_{ijkd} + \boldsymbol{\delta}_d + \boldsymbol{\theta}_{dr} + \boldsymbol{\theta}_t + \boldsymbol{e}_{ijkd} \quad [1]$$

where Y_{ijkd} refers to the health of child *i* of household *j* in cluster *k* of district *d*. Diarrhoea prevalence, weight-for-height, weight-for-age z-score, and ARI are the indicators of child health and the outcome variables of interest. WQ_{jkd} is a dichotomous variable that denotes POC water quality (presence of *E. coli*) in household *j* in cluster *k*. H_{ijkd} is a vector for socio-demographic controls at the individual, household, and community levels. δ_d controls district-level characteristics as explained in the data section. θ_{dr} controls for district-times-rural fixed effects. As rural and urban areas may have different characteristics within each district in terms of socioeconomic and health status, we include district-times-rural fixed effects to account for any region-specific unobserved factors. θ_t is the time-fixed effects, while e_{ijkd} is the error term that elucidates the variation of Y_{ijkd}. We cluster standard errors at the MICS cluster level.

Since POC drinking water is influenced by household decisions and behaviours, our model may suffer from an endogeneity issue.

[7] The following precise question was asked: Could you please provide me with a glass of water that members of your households usually drink?.
[8] The following precise questions were asked: 'What source was this water collected from?' and 'Can you please show me the source of a glass of drinking water so that I can take a sample from there as well?'
[9] Flush toilets reduced child weight loss, offer non-health welfare (e.g. time-saving), and augments satisfaction [23,62].

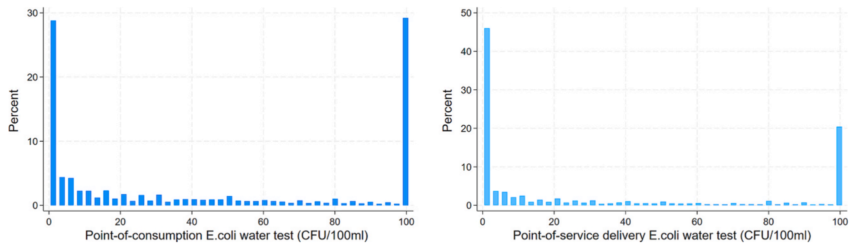

Fig. 1. Point of consumption and source water quality status (*E. coli* [CFU/100 ml]).

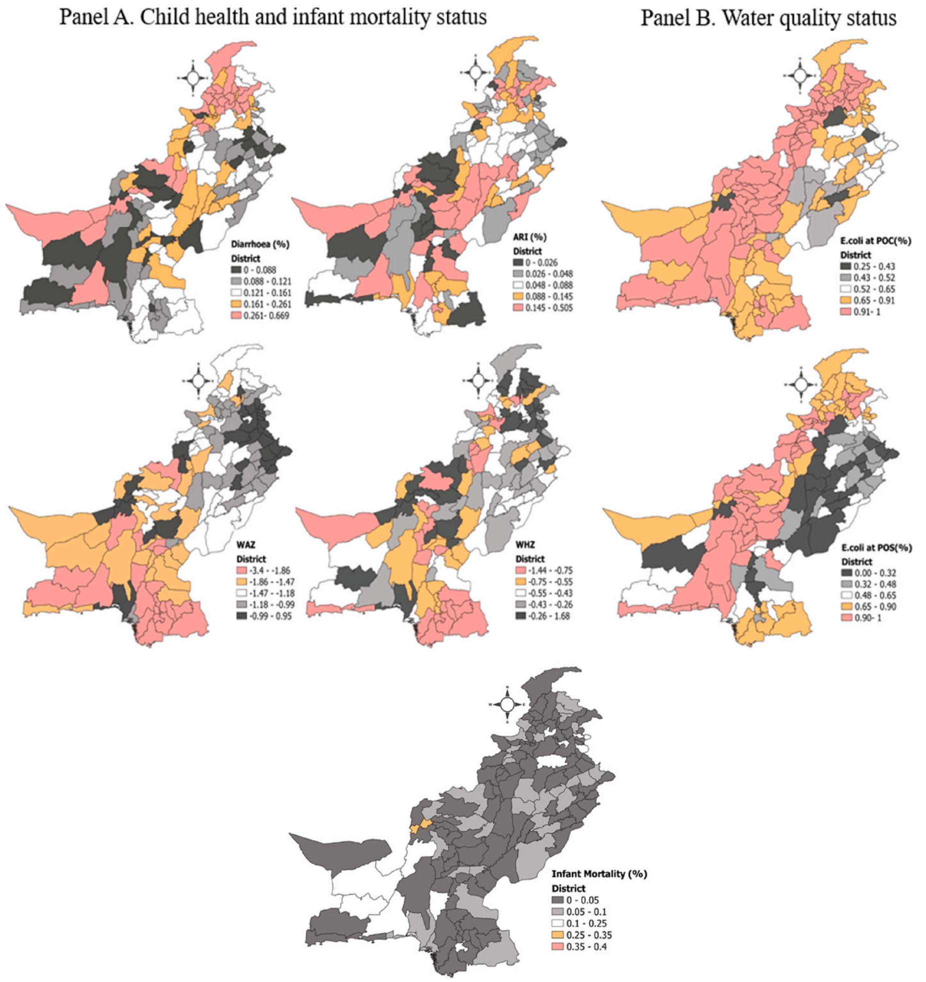

Fig. 2. Microbial water contamination and child health and infant mortality status at each district in 2017–2021.

For example, some unobserved factors (e.g. belief in safe sanitation access, awareness about clean water, averting behavior and hygiene practices) that we cannot capture fully in our estimation, could be correlated with household water quality. For instance, household's awareness of their POC poor water quality might engage them in more avoidance behavior (e.g., boiling water and using filters), which mitigates the potential health impacts of contamination, leading to underestimation of the observed negative impact of *E. coli* water contamination on health. Conversely, unobservable factors could lead to the overestimation of the negative impact of poor water quality if households with worse water quality have lower awareness about clean water, meaning lower avoidance behavior.

To circumvent such issues, we control as many covariates as possible, and we use 2SLS with an instrumental variable approach. The following equation is the first stage of the 2SLS estimation of equation (1):

$$\mathrm{WQ}_{jkd} = \gamma_0 + \gamma_1 \mathrm{SWQ}_{kd} + \gamma_4 \mathbf{H}_{ijkd} + \boldsymbol{\delta}_d + \boldsymbol{\theta}_{dr} + \boldsymbol{\theta}_t + \boldsymbol{\nu}_{jkd} \quad [2]$$

where the excluded instrumental variable is community-level POS water contamination, indicated as SWQ_{kd}.This measure is less likely to be correlated with individual, household-level decisions, behaviours, and other factors than POC water quality. As we discussed, in the MICS, the POS water quality data was measured at the household level because households use water from different sources. Each household cannot change the quality of the water at the source, however, there could be some possibility of endogeneity because households may be able to select their water sources based on several factors such as socioeconomic status, infrastructure, and proximity to clean water sources and this choice could be influenced by unobserved characteristics. For this reason, we use community-level POS water contamination status, calculated by measuring the leave-one-out mean as an instrument to create an exogenous measure of water quality for each household[10]. This ensures that the POS water quality is based on factors beyond the household's control, thereby reducing the potential for bias introduced by household-level characteristics in water source selection. Therefore, relying on this IV, we attempt to circumvent the endogeneity associated with household-level factors and provide a more exogenous measure of water quality. Moreover, because the average community-level POS water contamination most likely affects children's health mainly through drinking water quality, the variable could satisfy the exclusion restriction, although there is no direct way to test it.

Community-level POS water quality (except self), however, may not be completely exogenous due to the possibility of placement bias i.e., households living in the areas with better socio-economic status, may have good access to clean water and health facilities. To mitigate the potential impact of spatial correlation, we incorporate a comprehensive set of controls at the household, individual, and community levels. Specifically, we control for possible community-level characteristics that may be correlated with IV and also affect outcomes[11] such as community-level average open defecation behaviour, piped water access, and land and housing ownership. These characteristics capture the localized variations in socioeconomic, demographic, and environmental factors that may affect both POS water quality and health outcomes.

4. Results

4.1. Descriptive statistics

In Table 1, we compare child health and growth outcomes among households with POC drinking water with and without *E. coli* contamination. Table 1 shows that diarrhoea prevalence is higher by 5 % among children from households with contaminated drinking water than households with E. coli-free water. The average weight-for-height and weight-for-age z-scores are lower among children from households exposed to *E. coli* contamination than among those using uncontaminated water. On average, the weight-for-height is −0.43 for children exposed to contaminated water and −0.35 for those who drink water free of *E. coli*. Similarly, the weight-for-age of children with contaminated water is −1.33, while that of children with uncontaminated water is −1.11, and the difference is statistically significant. Further, the ARI prevalence is also higher by 1.3 % among children exposed to contaminated water than among households using uncontaminated water. However, no significant difference is observed in infant mortality among households with either contaminated or uncontaminated water.

4.2. Main results

Estimates of the impact of POC *E. coli* in water on child diarrhoea, weight-for-height, weight-for-age, and ARI prevalence are presented in Table 2. The estimated coefficients of contaminated water on weight-for-height and weight-for-age are negative and statistically significant. The results demonstrate that having *E. coli* in POC water significantly decreases the child's weight-for-age z-score by 0.051. POC water contamination is negatively associated with the child's weight-for-height z-score although the coefficient is insignificant with a *t*-value of 1.22. Due to the possible correlation between e_{ijkd} *and* WQ_{jkd}, however, the ordinary least squares (OLS) estimates of β1 may suffer from endogeneity bias.

To circumvent this problem, we estimate the same model using community-level POS water quality (except self) as IV for household-level POC water quality. Panel A of Table 3 shows the first-stage estimation results. The results show a strong positive correlation between POC water quality and IV, and diagnostic tests provide evidence of the validity of IV. For all outcomes, the null hypothesis for under-identification is rejected based on the LM version of the Kleibergen-Paap rk (rank) statistic. The null hypothesis of weak IV is rejected for all regressions using the Kleibergen-Paap rank Wald statistic, indicating that our instrument is not weakened by the inclusion of an extensive set of covariates. The reduced form results are shown in Appendix Table A3.

In Panel B of Table 3, we estimate β_1 by applying 2SLS that captures the impact of POC water quality on child health outcomes. Column 1 shows that the probability of diarrhoea in children increases by 11 percentage points in households with contaminated drinking water compared to those using uncontaminated water; estimated at the 17.8 % sample mean, this translates to a 61.8 % increase in diarrhoea risk. The results for weight-for-height and weight-for-age as alternative outcomes are shown in columns 2 and 3. Consistent with childhood diarrhoea triggering weight loss, particularly in the short run, we find that POC-contaminated drinking

[10] For community-level POS water quality, we excluded the particular household under consideration and measured the community-level average POS water contamination status (leave-one-out mean), following the ideas of [63,64].

[11] Koolwal and Van de Walle (2013) [65] argued that geographic averaging with rigorous community-level controls in regression is a defendable approach while using geographic instrumental variables.

Table 1
Mean comparison tests for child health outcomes with and without *E. coli* in household drinking water.

Household POC water quality (*E. coli*)				
Variables	Contaminated	Uncontaminated	Total	*t*-test
Diarrhoea	0.188	0.138	0.178	−0.050[b]
Weight-for-height	−0.439	−0.350	−0.421	0.089[b]
Weight-for-age	−1.333	−1.114	−1.287	0.218[b]
ARI	0.099	0.087	0.097	−0.013[a]
Observations	10,718	2885	13,603	
Infant mortality	0.046	0.047	0.046	−0.0004
Observations	4318	1202	5520	

Notes: Authors' calculations. Test for the mean comparison between households consuming water uncontaminated and contaminated with *E. coli*.
[a] $p < 0.10$, $^{**}p < 0.05$.
[b] $p < 0.01$.

Table 2
OLS estimates of the effects of drinking water quality on child health. N = 13,603.

Dependent variables	Diarrhoea	Weight-for-height	Weight-for-age	ARI
	(1)	(2)	(3)	(4)
Water quality (1 = E.coli present)	0.010	−0.039	−0.051[a]	0.004
	(0.009)	(0.032)	(0.031)	(0.007)
Controls				
Individual and Household Characteristics	YES	YES	YES	YES
Community Characteristics	YES	YES	YES	YES
District Characteristics	YES	YES	YES	YES
District x Rural FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES

Notes: Clustered standard errors (robust) at the MICS cluster level are in parentheses; $^{***}p < 0.01$, $^{**}p < 0.05$.
[a] $p < 0.1$.

Table 3
Effects of drinking water quality on child health (IV-2SLS). N = 13,603.

Regression stage	Water quality (1 = E.coli present)			
	(1)	(2)	(3)	(4)
Panel A. First-stage regression				
Community-level POS water contamination (except self)	0.198[a] (0.0161)	0.197[a] (0.0161)	0.196[a] (0.0162)	0.188[a] (0.0162)
Control variables				
District x Rural FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Individual Characteristics	YES	YES	YES	YES
Household Characteristics		YES	YES	YES
Community Characteristics			YES	YES
District Characteristics				YES
Kleibergen-Paap rk LM statistic (p-value)	120.26 (0.000)			
Kleibergen-Paap (KP) rk Wald *F*-statistic	134.6			
Dependent variables	Diarrhoea	Weight-for-height	Weight-for-age	ARI
	(1)	(2)	(3)	(4)
Panel B. Second-stage regression				
Water quality (1 = *E. coli* present)	0.110[b]	−0.418[b]	−0.397[b]	0.039
	(0.053)	(0.202)	(0.188)	(0.043)

Notes: Panel A shows the first stage of two-stage least squares, where the excluded instrumental variable is community-level source water contamination, and the endogenous explanatory variable is household POC microbial water quality. Panel B shows results for the second stage of 2SLS. The endogenous variable is instrumented by community-level source water contamination. The Kleibergen-Paap rk LM test strongly rejects the null hypothesis of under-identification. Further, the null hypothesis that IV is weakly correlated with endogenous variable is strongly rejected by the Kleibergen-Paap rk Wald *F* statistic. The values also exceed Stock-Yogo weak ID test critical values. A set of controls similar to Panel A column 4 applies. Clustered standard errors (robust) at the MICS cluster level are in parentheses.
[a] $p < 0.01$.
[b] $p < 0.05$, $^{*}p < 0.1$.

Table 4
The effect of water quality on infant mortality (per 1000 children). N = 5520.

Variables	OLS	OLS	IV- First stage	IV- Second stage	IV- Second stage	IV- Second stage
	IMR	IMR	POC Water quality	IMR	IMR	Premature birth
	(1)	(2)	(3)	(4)	(5)	(6)
Water quality (1 = E.coli present)	8.035	7.754		7.674	2.345	−0.056
	(7.33)	(7.45)		(37.74)	(37.96)	(0.06)
Community-level POS water contamination (except self)			0.205[a]			
			(0.020)			
Mean of dependent variable	46.92	46.92	0.78	46.92	46.92	0.12
Premature birth	YES	–	YES	YES	–	–
Controls	YES	YES	YES	YES	YES	YES
KP rk Wald *F*-statistic			103			103

Notes: The dependent variable is infant mortality rate (IMR) in columns 1,2,4, and 5. Premature birth takes 1 if the child was born before 37 weeks of pregnancy. The individual level controls include child gender, birth order, child age in months, premature birth, prenatal care during pregnancy, household head gender, and mother's education. A set of household, cluster, and district level controls, and fixed effects similar to Table 3 panel B applies. Clustered standard errors (robust) at MICS cluster level in parentheses.
[a] $p < 0.01$, **$p < 0.05$, *$p < 0.1$.

water is associated with a decrease in children's weight-for-height by 0.41 standard deviations and their weight-for-age by 0.39 standard deviations. In contrast, column 4 indicates the insignificant and moderate effects of POC-contaminated water on the probability of ARI risk.

One potential concern could be the possibility of survival bias, which may mask the mortality effect in our analysis presented in Table 3. The infants severely affected by contaminated water, which possibly led to death, may not be included in the sample. This exclusion could underestimate the true effect of water quality on child health for the surviving children in our study. In Table 4, we test this hypothesis and examine the effects of contaminated water on IMR. We estimate the following model similar to equation (1).

$$\mathbf{Y}_{ijkd} = \beta_0 + \beta_1 \mathrm{WQ}_{jkd} + \beta_2 \mathbf{X}_{ijkd} + \boldsymbol{\delta}_d + \boldsymbol{\theta}_{dr} + \boldsymbol{\theta}_t + \boldsymbol{\mu}_{ijkd} \quad [3]$$

Where i is the index of live births and the dependent variable is Y_{ijkd} is an individual-level mortality indicator for infant death. We scaled this because the coefficients indicate impacts on the infant deaths per thousand [51]. WQ_{jkd} is our key exposure variable. X_{ijkd} denote controls at the individual, household, and cluster levels. The individual level controls include the child's gender, birth order, child age in months, probability of premature birth, prenatal care during pregnancy, and mother's education. A set of household, cluster, and district-level characteristics, and geographic and time fixed effects are added similar to equation (1).

The results in Columns 1 and 4 of Table 4 report the main OLS and IV estimates with an extensive control set. The *E. coli* water contamination is found to be insignificantly associated with IMR.[12] Column 3 presents the first-stage estimations. The results suggest that the possibility of survival bias is unlikely in our case. The waterborne illnesses may have a spectrum of severity. While it can be fatal in some cases, particularly for weaker infants, other cases might be milder, and would not necessarily lead to death in young infants with strong immune systems. These children also have less environmental exposure, making them less susceptible to severe waterborne illness.

Given the possibility that poor water quality may have an indirect impact on the IMR through its effects on an infant's premature birth, in column 6 of Table 4 we observe an insignificant effect of water quality on the probability of premature birth. Moreover, the impact of POC-contaminated water on IMR remains statistically insignificant and quantitatively slightly smaller as reflected in columns 2 and 5 of Table 4 if the dummy for a child's premature birth is not included in the model. This suggests that the variable of premature birth absorbing a part of the water quality effect is less likely in this case.

The IV estimates of the main results indicate that the effect of water contamination on child health indicators and IMR are larger than those from OLS estimations and are statistically significant at a 5 % level except for ARI and IMR. Although the estimated coefficients of the OLS estimates are insignificant, the direction of the impact is the same as that of the IV estimates, implying the robustness of our results. This may suggest the presence of bias from the endogeneity in OLS estimates, possibly due to unobserved factors correlated with household water quality and health outcomes.

4.3. Robustness checks

We conduct an array of robustness checks to document the validity and stability of our results. First, we examine whether the estimated 2SLS coefficients remain robust after excluding extensive covariates. In the main models, we control for variables related to the individual, household, community, and district-level characteristics, geographic location fixed effects, and time-fixed effects. Fig. 3

[12] We restrict the sample to children born within the last 2 years before the survey, assuming that water quality remained constant for two years. Although the results are not shown, in a robustness check, we observe an insignificant effect of water quality on IMR using a sample of children born in the last year before the survey, assuming that water quality remained constant for one year.

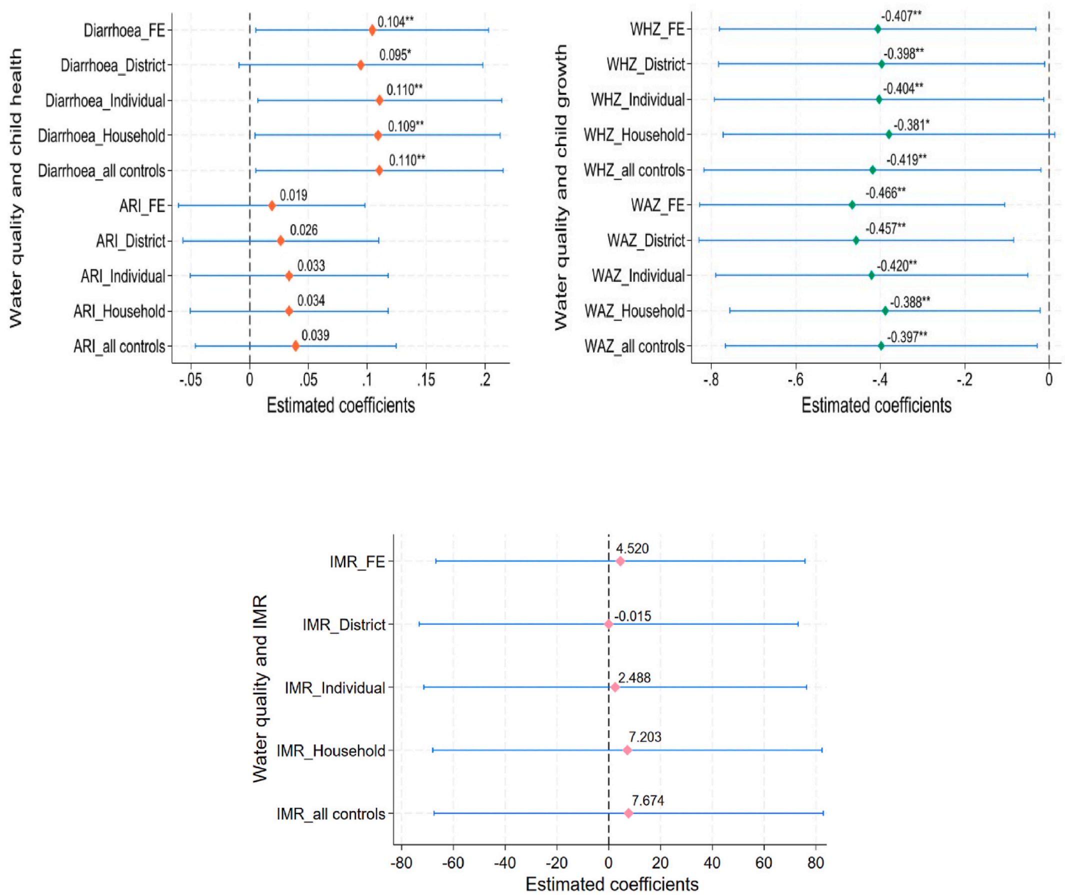

Fig. 3. Relationship between *E. coli* contamination and child health, growth, and infant mortality with gradual inclusion/exclusion of controls. The figure visualises the estimated coefficients from 5 regressions for each outcome variable in the IV framework. In the first regression for each outcome, all controls except year-fixed effects and district-times-rural fixed effects are excluded. The second regression incorporates district-level characteristics in addition to fixed effects. Next, in the third and fourth regression individual and household level controls are added. Lastly, cluster-level controls are added to all other controls.

presents the detailed IV estimations with gradual inclusion and exclusion of controls. In the initial estimations for each outcome, we exclude all controls except fixed effects for each outcome as shown in Fig. 3. Next, we gradually include district-level characteristics, and individual and household controls. In the last regression, we include all controls (similar to Panel B of Table 3). The results show that the coefficient of water quality remains largely steady despite the inclusion or exclusion of various covariates.

Second, we add several variables that could affect water quality in the first- and second-stage regressions in Tables 3 and 4. We include the means to extract water (motor, hand pump, and tube-well dummy), piped water dummy, water treatment status dummy, handwashing dummy, flush toilet dummy, and animal ownership. As these household facilities can potentially determine water quality, excluding them may cause an omitted variable bias, especially in the first-stage estimation. At the same time, however, these variables are endogenous, and thus, we exclude them from our main analyses. Appendix Table A4 and Appendix Table A5 show the first stage estimates in Panel A and IV estimates with and without these variables in Panel B. The results show that the coefficients are almost similar for the estimation results with and without the inclusion of potential water quality determinants, suggesting the robustness of the results.

Lastly, in Appendix Table A6, as a robustness check, we re-estimate our model by measuring the *E. coli* concentration as a binary indicator, above or below the median. We use this definition because the values of E.coli count above 100 [CFU/100 ml] were censored and coded as 101. Thus, we do not use water quality status as a continuous indicator. Using this alternate specification, our IV-2SLS estimations remain coherent with the main results. Overall, the robustness tests suggest that our findings are robust and consistent.

4.4. Differential effects of E. coli contamination

In the previous estimations, we assume that the effects of *E. coli* contamination are the same across all sample children. We further assess whether the impact of POC water quality varies across the different age groups as *E. coli* contamination may not similarly affect all children. The results presented in Table 5 show stark differences in the impact of contaminated water on child health among

Table 5
Differential effects of drinking water quality on child health by age group.

Variables	Aged 6 months and younger (IV)				Older than 6 months of age (IV)			
	Diarrhoea	Weight-for-Height	Weight-for-Age	ARI	Diarrhoea	Weight-for-Height	Weight-for-Age	ARI
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Water quality (1 = *E. coli* present)	0.047	−0.040	−0.217	0.006	0.122[a]	−0.454[a]	−0.376[a]	0.043
	(0.150)	(0.599)	(0.572)	(0.115)	(0.055)	(0.207)	(0.192)	(0.045)
Observations	1585	1585	1585	1585	12,018	12,018	12,018	12,018
F-Statistic	27.2				134.9			

Notes: The estimations in columns 1–4 show the water quality impact on children's health aged 0–6 months, while columns 5–8 show such impacts on children older than 6 months. A set of controls similar to Table 3 Panel B applies. During the initial six months, children are usually exclusively breastfed and, after the introduction of complementary feeding and partial weaning from the age of 6 months increases a child's exposure to a contaminated environment. Clustered standard errors (robust) at the MICS cluster level are in parentheses; ***$p < 0.01$.
[a] $p < 0.05$, *$p < 0$.

different age groups. The IV results in columns 1–4 show the impact of water quality on the health of children aged 6 months and younger, while columns 5–8 show such effects on children older than 6 months of age. The estimated coefficients in columns 5–8 indicate that the link between E. coli-contaminated water and the risk of diarrhoea incidence is statistically significant among children older than 6 months, with an average effect of 12.2 percentage points. Moreover, contaminated water is significantly associated with a 0.45 and 0.37 standard deviation decrease in weight-for-height and weight-for-age z-scores respectively, in children older than 6 months. However, the effect on ARI risk remains insignificant. Note that, as previously discussed, the possibility of survival bias is unlikely to be the major explanation for the insignificant impact on younger children's health outcomes because we observe insignificant effects of water contamination on IMR.

In contrast, the results in columns 1–4 show that water pollution does not affect children aged 6 months and younger. This is plausible because the introduction of complementary feeding and partial weaning from 6 months increases a child's exposure to a contaminated environment.[13] Children exclusively breastfed during the initial 6 months may be protected naturally from pathogen exposure through contaminated food and water, having less ARI, diarrhoeal morbidity, mortality [52,53], and growth failure [54].[14] While further exploring this channel, we conduct estimations based on exclusive breastfeeding practices in Appendix Table A7. We find an insignificant effect of water quality on child health indicators focusing on children 0–6 months who are either mostly or exclusively breastfed. However, the size of the coefficient, for the non-exclusively breastfed group suggests that they could be less protected from microbial water pollution than those who are completely breastfed. This finding is not puzzling and remains aligned with our age-specific results for infants 6 months and younger.[15] Based on the observed effects, we cannot be conclusive about the mechanism, however, breastfeeding seems to be one of the potential mechanisms.

In addition, we analyse whether the intake of vitamin A supplements influences the effects of E. coli-induced health risks. Vitamin A deficiency (VAD) remains a public health concern across the globe including in Pakistan, where its frequency is more than 51 %. According to previous studies, vitamin A (anti-infectious) is significant for gastrointestinal and respiratory epithelial regeneration and reduces immune disorders, mortality, and morbidity among children aged 6–59 months [55–57]. A periodic supplementation could protect children by boosting their immune systems from contaminated disease environments and related health risks [58].

To examine these hypotheses, we conduct a subsample analysis of the impact of water quality on child health based on vitamin A supplementation. In our analyses, the vitamin A variable takes 1 if the child aged 6–59 months was given vitamin A supplementation in the previous 6 months.[16] Table 6 shows the differential effects of drinking water quality on the health and growth of children with and without vitamin A supplementation. Columns 1–4 show such impact on the outcome variables for the non-vitamin A-supplemented group. The results in column 1 show that contaminated water is significantly associated with the risk of diarrhoea in the non-supplemented group indicating a negative impact of *E. coli* on health outcomes for children in this group. However, columns 2 and 3 show an insignificant effect of water quality on child growth suggesting that even if provided with clean water, growth indicators are not affected when they lack vitamin A supplements. For the vitamin A-supplemented group, the results in columns 5–7 show that water

[13] In terms of food intake for the previous day, the information was only solicited for children of up to 2 years at the time of the survey.
[14] In our sample around 43 % of children were exclusively breastfed, and 56 % were predominantly breastfed among sample children under six months of age. Consistent with WHO recommendations, recent studies have found that the median duration of exclusive breastfeeding for children is 6 months [66].
[15] A similar trend is observed for the predominantly breastfed children (which allows for the intake of vitamin drops or syrups, mineral supplements, or prescribed medicines but no food-based fluids except fruit juice and sugar water).
[16] The vitamin A supplementation by Nutritional International program in Pakistan, supported by WHO, is conducted twice a year during national immunization days aiming to reach all the children aged 6–59 months. Although, we controlled several factors in our regressions. We cannot deny the potential endogeneity of vitamin A intake due to unobserved factors. However, MICS data shows that the treated and control groups are similar in terms of socio-economic status, maternal education, gender, and geographical location suggesting that the program did not disproportionally focus on specific socio-economic groups and effectively covered a broad demographic [39].

Table 6
Differential effects of drinking water quality on child health depending on Vitamin A supplementation.

Variables	Vitamin A Non- Supplemented group (IV)				Vitamin A Supplemented group (IV)			
	Diarrhoea	Weight-for-Height	Weight-for-Age	ARI	Diarrhoea	Weight-for-Height	Weight-for-Age	ARI
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Water quality (1 = *E. coli* present)	0.163[b]	−0.122	0.251	0.102	0.134[b]	−0.797[a]	−0.908[a]	0.040
	(0.085)	(0.322)	(0.303)	(0.063)	(0.073)	(0.280)	(0.271)	(0.063)
Observations	5265	5265	5265	5265	6297	6297	6297	6297
F-Statistics	61.74				81.45			

Notes: The results in columns 1–4 show the effects of *E. coli* water contamination on the outcome variables for the vitamin A-supplemented group, while columns 5–8 show such effects on the non-supplemented group from four provinces Punjab, Sindh, Baluchistan, and KPK in Pakistan. A set of controls similar to Table 3 Panel B applies. Clustered standard errors (robust) at MICS cluster level are in parentheses.
[a] $p < 0.01$, $^{**}p < 0.05$.
[b] $p < 0.1$.

quality significantly affects a child's diarrhoea, weight-for-height, and weight-for-age z-scores. It might be possible due to baseline vitamin A deficiencies, underlying nutritional deficiencies, and pre-existing health conditions to affect mainly child growth indicators. This may suggest that the combination of clean water access and vitamin A provision might work synergistically to improve growth indicators and diarrhoeal outcomes. The effect of water quality on ARI remains insignificant in both groups.

5. Discussion

This study investigates the impact of household water quality on the incidence of childhood diarrhoea, weight-for-height, weight-for-age, and ARI in Pakistan using MICS data. Globally, only a few datasets include bacteriological water quality indicators at the national scale. MICS is a unique data set that allows us to directly examine the impact of POC water quality on children's growth and health. Moreover, previous studies have largely ignored the endogeneity of POC drinking water quality, which might have caused bias in the estimates. We addressed these concerns by employing the two-stage least squares method and tried to control for possible observed factors. We find that poor drinking water quality significantly and negatively affects child growth and increases diarrhoea incidence. However, contaminated water insignificantly affects ARI risk and IMR.

Our findings suggest that POC water contamination elevates the risk of malnutrition and diarrhoea. Water contamination primarily affects gastrointestinal health, hence the negative impact of *E. coli* contamination on diarrhoea is consistent with previous studies (e.g. Usman et al., 2019, Rahman et al., 2021; Khan et al., 2022) [16,22,23], and we add evidence to the growing literature on the negative impact of *E. coli* contamination on weight-for-height, weight-for-age, and ARI by employing rigorous empirical estimations. We observe an insignificant effect of contaminated water on ARI, possibly because diarrhoea, weight-for-height, and weight-for-age are directly linked to the gastrointestinal system, whereas the link between ARI and E.coli is less direct.[17] Our results also reveal insignificant effects of microbial water quality on IMR, possibly due to infants' limited exposure to water pollution.

Age-based sub-sample analyses demonstrate that contaminated water significantly affects child health indicators, particularly when a child turns above 6 months. A reason for this could be that after 6 months, children consume large quantities of water and are at a higher risk of being infected through contact with contaminated environment. Infants (6 months and younger) may be naturally protected by breastfeeding and their limited exposure to water pollution, we did not observe a significant negative impact of contaminated water on their health. Additionally, we investigate the exclusive breastfeeding channel and observe an insignificant effect of water quality on the health of children who are either mostly or exclusively breastfed. This finding is consistent with our results for children 0–6 months as demonstrated in Table 5. Please note that we are not conclusive about the mechanism, however, breastfeeding can be a potential mechanism that needs to be further explored. Furthermore, an insignificant effect of poor water quality on IMR in the main analysis indicates that survival bias is unlikely to be the major explanation for the insignificant effect on health outcomes in this age-specific differential analysis.

We examine the differential impacts of water quality depending on the provision of micronutrient supplements to children. We show that while clean water access improves child health, it does not help child growth in the non-supplemented group, possibly because Vitamin A deficiency is a bottleneck for a child's growth. On the contrary, when children are supplemented with vitamin A, clean water significantly reduces diarrhoea and improves weight-for-height and weight-for-age z-scores. It might be possible due to baseline vitamin A deficiencies, underlying nutritional status, and pre-existing health conditions of children. The clean water access and vitamin A supplementation may work as complements to improve children's growth and health outcomes. The observed results highlight the potential benefits of combined intervention – clean water access and vitamin A supplementation.

[17] Certain strains of *E. coli* may develop resistance to multiple antimicrobial drugs (MDR), leading to MDR *E. coli* infections. Contaminated water could harbor some strains that contain antibiotic-resistance genes and virulence factors, which may contribute to the severity of various infections, including respiratory tract infections.

Despite our contributions, this study has some limitations. Firstly, water quality was measured at a single point in time; therefore, we could only rely on cross-sectional variations to estimate the health impact of water quality. Undernutrition reflects the accumulated effects of nutritional deficiency; therefore, a longitudinal study on the water quality effects on chronic child undernutrition should be conducted in the future. Secondly, community-level POS water quality (except self) may not be completely exogenous. To mitigate the potential concern of spatial correlation, we attempt to incorporate an extensive set of controls and several community-level characteristics to capture the shared characteristics of communities that may affect both water quality and health outcomes. Due to data limitations, however, we could not control for other community-level spatial variables (e.g. distance to water sources and proximity to pollution sources) in our model. Thirdly, we do not have information on the seasonal variability of water quality and *E. coli* levels; thus, we cannot explore whether rainy or dry seasons affect water contamination and child health differently.

6. Conclusion

Pakistan is one of the countries that are susceptible to respiratory infections, diarrhoea-related mortality, malnutrition, and infant mortality rate (IMR) in South Asia. This study examines the impact of microbial water contamination on childhood diarrhoea, weight-for-height, weight-for-age, and IMR using data from the latest waves of MICS in Pakistan. Our key findings indicate that the presence of *E. coli* in POC water, which is generally used for drinking, significantly decreases a child's weight-for-height and weight-for-age and increases the risk of diarrhoea, although it does not significantly affects ARI incidence and IMR.

From our findings, we can extrapolate that the effect of POC microbial water contamination on child health is not trivial and highlight certain policy recommendations. First, financial allocations should be diverted towards safely managed drinking water provision particularly in unserved areas to improve people's health and to reduce costs attributed to avoidance behaviours. For instance, the establishment of waste-management treatment plants and sewerage schemes may contribute to improving children's health by reducing water contamination. Second, families with children should be realised the importance of a clean environment for their growing children, and exclusive breastfeeding can be beneficial. In Pakistan, currently, only approximately 48.4 % of women exercise exclusive breastfeeding [59]. The promotion of breastfeeding and vitamin A supplementation would directly impact child health and indirectly, through the combined intervention of clean water access and vitamin A supplementation.

Considering that the issue of poor child health is getting severe possibly due to recent massive floods in the country negatively affecting water safety and sanitation systems [19], it is necessary to conduct longitudinal analyses and observe the long-term impacts of water contamination as well as how infrastructure and hygiene practices can protect child health.

Disclosure statement

No potential conflict of interest was reported by the authors.

CRediT authorship contribution statement

Ammazia Hanif: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Yuko Nakano:** Writing – review & editing, Validation, Supervision, Methodology. **Midori Matsushima:** Writing – review & editing, Validation, Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data used in this article is obtained from the MICS program website (https://mics.unicef.org/surveys), which is publicly archived and freely available for download.

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Appendix

Appendix Table A1
Summary Statistics for the whole sample (N = 13,603)

Variables	Mean	Std. Dev.	Min	Max
HH water quality (1 = *E. coli* present)	0.787	0.408	0	1
Diarrhoea	0.178	0.382	0	1
Weight-for-height z-score	−0.421	1.344	−5	5
Weight-for-age z-score	−1.287	1.307	−5.99	4.94
ARI	0.097	0.296	0	1
HH Demographic and Socioeconomic				
Child gender (1 = male)	0.516	0.499	0	1
Child age (month)	30.169	17.026	0	60
Children under 5 years	2.283	1.324	1	10
HH head gender (1 = male)	0.922	0.268	0	1
Maternal education	0.401	0.490	0	1
Maternal age	30.110	6.323	15	49
Maternal smoking status	0.021	0.143	0	1
Number of HH members	9.158	4.794	2	42
Wealth index quantile (1 = poorest, …, 5 = richest)	2.661	1.364	1	5
Cooking place (1 = outdoor)	0.283	0.450	0	1
Flush toilet	0.628	0.483	0	1
Animal ownership	0.533	0.498	0	1
Year	2018	0.781	2017	2021
Community Characteristics				
No toilet facility	0.186	0.248	0	1
Piped water access	0.243	0.287	0	1
House ownership	0.857	0.239	0	1
Land ownership for agriculture	0.341	0.343	0	1
Residential area (1 = rural)	0.779	0.414	0	1
District Characteristics				
Log of population	5.792	1.474	1.107	10.847
Malaria prevalence rate	0.006	0.008	0.00005	0.045
Night-time light (nanoWatts/cmˆ2/sr)	1.643	6.229	0.208	55.690
Rainfall (mm)	518.061	382.577	44.860	1556.31
Temperature (Celsius)	31.023	6.202	8.902	41.979
Slope (degree)	7.003	8.769	0.281	34.355

Appendix Table A2
Summary Statistics for sub-sample for the analysis of infant mortality rate (N = 5520)

Variables	Mean	Std. Dev.	Min	Max
HH water quality (1 = *E. coli* present)	0.782	0.412	0	1
Infant mortality	0.046	0.211	0	1
HH Demographic and Socioeconomic				
Child gender (1 = male)	0.513	0.499	0	1
Child age (month)	11.190	6.732	0	23
Birth order	2.240	0.896	1	4
Premature birth	0.124	0.330	0	1
Prenatal care during pregnancy	0.749	0.433	0	1
HH head gender (1 = male)	0.920	0.270	0	1
Maternal education	0.430	0.495	0	1
Maternal age at birth ((1 = <20, …, 3 = 35+)	2.058	0.472	1	3
Maternal smoking status	0.020	0.141	0	1
Number of HH members	9.177	4.776	2	42
Wealth index quantile (1 = poorest, …, 5 = richest)	2.716	1.361	1	5
Cooking place (1 = outdoor)	0.280	0.449	0	1
Flush toilet	0.656	0.474	0	1
Animal ownership	0.539	0.498	0	1
Year	2018	0.781	2017	2021
Community Characteristics				
No toilet facility	0.176	0.240	0	1
Piped water access	0.243	0.286	0	1
House ownership	0.858	0.236	0	1
Land ownership for agriculture	0.346	0.345	0	1
Residential area (1 = rural)	0.782	0.412	0	1
District Characteristics				
Log of population	5.857	1.428	1.107	10.847

(*continued on next page*)

Appendix Table A2 (*continued*)

Variables	Mean	Std. Dev.	Min	Max
Malaria prevalence rate	0.005	0.008	0.00005	0.045
Night-time light (nanoWatts/sr/cmˆ2)	1.586	5.942	0.208	55.690
Rainfall (mm)	532.323	384.520	44.860	1556.319
Temperature (Celsius)	30.732	6.284	8.902	41.979
Slope (degree)	7.055	8.927	0.281	34.355

Appendix Table A3
Reduced-form results

Dependent Variable					
Variables	Diarrhoea	Weight-for-height	Weight-for-age	ARI	IMR
	(1)	(2)	(3)	(4)	(5)
Community source water quality (except self)	0.020**	−0.078**	−0.074**	0.007	1.573
	(0.010)	(0.037)	(0.035)	(0.008)	(7.861)
Controls (all)	YES	YES	YES	YES	YES
N	13,603	13,603	13,603	13,603	5520

Notes: Clustered standard errors (robust) at the MICS cluster level are in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Appendix Table A4
Sensitivity analysis with and without WASH infrastructure and hygiene-related variables for IV estimates of the impact of water quality on health.

Panel A. First-stage regression	Without water quality determinant variables		With water quality determinant variables	
	Water quality (1 = *E. coli* present)			
	(1)		(2)	
Community-level source water quality	0.188*** (0.0161)		0.189*** (0.0161)	
Panel B. Second-stage regression	Diarrhoea		Weight-for-height	
	(1)	(2)	(3)	(4)
Water quality (1 = *E. coli* present)	0.110**	0.112**	−0.418**	−0.412**
	(0.053)	(0.053)	(0.203)	(0.202)
	Weight-for-age		ARI	
	(5)	(6)	(7)	(8)
	−0.397**	−0.393**	0.039	0.038
	(0.188)	(0.188)	(0.043)	(0.043)
Piped water		YES		YES
Motorized pump/borehole		YES		YES
Tube well		YES		YES
Water treatmen		YES		YES
Handwashing		YES		YES
Flush toilet	YES	YES	YES	YES
Animal ownership	YES	YES	YES	YES

Notes: Access to flush toilets and animal ownership are included as controls in the main model. Columns 1,3,5 and 7 in panel B present the results of 2SLS estimations similar to Table 3. The regression results presented in Columns 2, 4, 6 & 8 control for potential factors that can significantly affect microbial water quality. Clustered standard errors (robust) at the MICS cluster level are in parentheses.***$p < 0.01$.**$p < 0.05$, *$p < 0.1$. N = 13,603.

Appendix Table A5
Sensitivity analysis with and without WASH infrastructure and hygiene-related variables for IV estimates of the impact of water quality on infant mortality.

Panel A. First-stage regression	Without water quality determinant variables	With water quality determinant variables
	Water quality (1 = *E. coli* present)	
	(1)	(2)
Community-level source water quality	0.204*** (0.020)	0.204*** (0.020)
	IMR	
Panel B. Second-stage regression	(1)	(2)
Water quality (1 = *E. coli* present)	7.673 (37.742)	6.258 (37.814)
Piped water		YES
Motorized pump/borehole		YES
Tube well		YES
Water treatment		YES
Handwashing		YES
Flush toilet	YES	YES
Animal ownership	YES	YES

Notes: Panel A presents first-stage and second-stage 2SLS results with and without water quality determinants on the analogy of Appendix Table A4. Clustered standard errors (robust) at the MICS cluster level are in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$. N = 5520.

Appendix Table A6
The IV estimates the impact of drinking water quality (measured above or below median [CFU/100 ml]) on child health and infant mortality.

Variables	Diarrhoea	Weight-for-height	Weight-for-age	ARI	IMR
	(1)	(2)	(3)	(4)	(5)
Water quality (1 = *E. coli* above median)	0.097**	−0.368**	−0.350**	0.034	6.988
	(0.047)	(0.178)	(0.166)	(0.038)	(34.40)
F-Statistic	142.5	142.5	142.5	142.5	106.3
N	13,603	13,603	13,603	13,603	5520

Notes: The main explanatory variable is POC water quality measured as a binary indicator and takes 1 if the E.coli count in water is above the median value of 26 and 0 otherwise. A set of controls similar to Table 3 panel B applies. Clustered standard errors (robust) at the MICS cluster level are in parentheses; ***$p < 0.01$.**$p < 0.05$, *$p < 0.1$.

Appendix Table A7
Differential effects of drinking water quality on child health by exclusive breastfeeding for children 0–6 months

Variables	Exclusively breastfed (IV)				Non-exclusively breastfed (IV)			
	Diarrhoea	Weight-for-Height	Weight-for-Age	ARI	Diarrhoea	Weight-for-Height	Weight-for-Age	ARI
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Water quality (1 = *E. coli* present)	−0.115	0.808	0.306	−0.195	0.330	−0.519	−0.833	−0.0739
	(0.200)	(0.768)	(0.691)	(0.157)	(0.298)	(1.149)	(1.180)	(0.232)
Observations	621	621	621	621	908	908	908	908
F-Statistic	13.51				7.53			

Notes: The estimations in columns 1–4 show the water quality impact on children's health aged 0–6 months who were exclusively breastfed, while columns 5–8 show such impacts on non-exclusively breastfed children. A set of controls similar to Table 3 Panel B applies. Clustered standard errors (robust) at the MICS cluster level are in parentheses; ***$p < 0.01$, **$p < 0.05$, *$p < 0$.

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